

Dataplex's data lineage feature

Previously, you learned that a data source is the place where data is generated. For many organizations, managing multiple data sources can be a complex undertaking, usually requiring extensive efforts to trace data back from its point of origin, to its current state. In this reading, you'll learn more about the process of data tracing, and how you can accomplish it using Dataplex's lineage feature.

Data lineage tracing

Data tracing includes tracking specific metadata that provides insight into the path the data has taken throughout its entire lifecycle. More specifically, data lineage tracing starts with tracking where the data comes from, including where it's passed to, and what transformations were applied to it. Services like Dataplex include built-in data lineage features that create graphs using your dataset's metadata.

Importance of data lineage

Data lineage serves several purposes beyond finding previous versions of data. Data lineage tracing assists with understanding how data is sourced and transformed, tracing errors in data and data input, and enabling change management.

Data lineage can also support regulatory compliance and privacy regulations. It can help you map data access points, and understand how data has been modified over its lifecycle by providing a record of these changes. Data lineage can also be used for audits and summary reports.

Data lineage graph

Dataplex creates a data lineage graph to visualize the changes in data. The data lineage model contains three types of information: process, run, and event.

Process

A data process is a transformation operation that was performed on a set of data. A data process may include revising or manipulating the data, including normalizing the dataset, summarizing the data, or sorting and indexing the data. A data process may also include combining or separating datasets.

Run

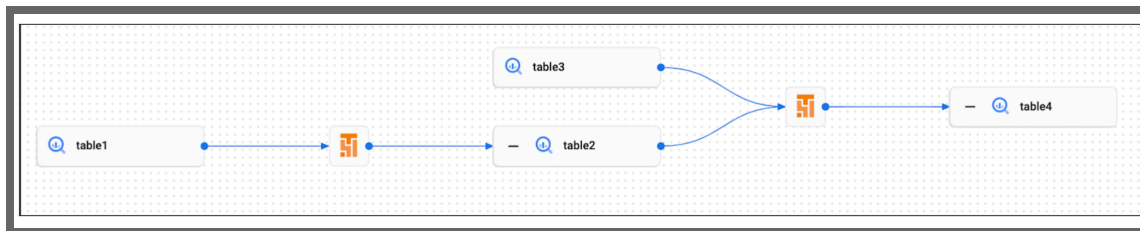
A run is a record of when the process was executed. Each process can have one or more runs. The run details include the start and end time of the process, and any other details determined by the organization.

Event

An event is what happens when data transformation takes place. The data transformation event occurs when data moves from a source to its final destination. An event will also include information like a link to the entry source.

Lineage graph components

The Dataplex lineage graph creates a visual representation of how data has been processed by different online services. The graph creates a rectangle icon for each version of a dataset. It also includes icons to indicate the service used to transform the data. Arrows indicate movement and any merging data.



The Dataplex lineage graph illustrates where table1 was processed by Cloud Composer and transformed into table 2. Icons representing Cloud Composer and BigQuery are common to the Dataplex data lineage graphics, and indicate which services processed the data.

Key takeaways

Dataplex's lineage feature enables you to track and visualize changes made to your datasets. It uses your dataset's existing metadata to track any changes or manipulations to the data, and creates a log to note any changes to the dataset. Visualization tools create a map of how data is transformed, and how it has reached its current state. This feature can be especially useful for editing, compliance, and error detection.

Resources for more information

To learn more about how Dataplex's data lineage graphs can support your work as a data analyst, check out these links:

- Google Cloud's blog provides information about understanding Dataplex's data lineage feature:
<https://cloud.google.com/blog/products/data-analytics/announcing-dataplex-data-lineage-in-ga>